

Home Assignment

1) Get the List of Open weight LLM from different service providers

a) Open ai

Model	Size	Best suited for
gpt-oss-120b	~117B total / 5.1B active	High-quality reasoning, agents, coding, complex tasks
gpt-oss-20b	~21B total / 3.6B active	Local deployment, lower latency, laptops/workstations
gpt-oss-safeguard-120b	~117B	Safety classification and policy reasoning
gpt-oss-safeguard-20b	~21B	Lightweight safety classification

b) Qwen

Generation	Model Variant	Primary Focus / Specialization	Available Sizes (Parameters)	Architecture	License
Qwen3	Qwen3	General Purpose LLM	30B-A3B, 235B-A22B (+ Dense variants)	MoE / Dense	Apache 2.0
	Qwen3-Coder	Code Generation & Agents	480B-A35B (+ smaller variants)	MoE / Dense	Apache 2.0
	Qwen3-VL	Vision-Language (Images/Video)	Multiple sizes (e.g., 8B, 32B, 72B)	Dense	Apache 2.0
	Qwen3-Omni	Omni-modal (Text, Audio, Vision, Speech)	Multiple sizes	Dense	Apache 2.0
Qwen2.5	Qwen2.5	General Purpose LLM	0.5B, 1.5B, 3B, 7B, 14B, 32B, 72B	Dense	Apache 2.0
	Qwen2.5-Coder	Code Generation & Debugging	0.5B, 1.5B, 3B, 7B, 14B, 32B	Dense	Apache 2.0
	Qwen2.5-Math	Mathematical Reasoning	1.5B, 7B, 72B (+ PRM/RM variants)	Dense	Apache 2.0
	Qwen2.5-VL	Vision-Language (High-Res Images/Video)	3B, 7B, 72B	Dense	Apache 2.0
	Qwen2.5-Omni	End-to-End Multimodal (Audio/Text/Vision)	7B	Dense	Apache 2.0

Generation	Model Variant	Primary Focus / Specialization	Available Sizes (Parameters)	Architecture	License
Qwen2	Qwen2	General Purpose LLM	0.5B, 1.5B, 7B, 57B-A14B, 72B	Dense / MoE	Apache 2.0*
	Qwen2-VL	Vision-Language	2B, 7B, 72B	Dense	Apache 2.0
	Qwen2-Math	Mathematical Reasoning	1.5B, 7B, 72B (+ PRM/RM variants)	Dense	Apache 2.0
	Qwen2-Audio	Audio Understanding & Speech	7B	Dense	Apache 2.0
Qwen1.5	Qwen1.5	General Purpose LLM	0.5B, 1.8B, 4B, 7B, 14B, 32B, 72B, 110B-A14B	Dense / MoE	Apache 2.0 / Tongyi*
Qwen1	Qwen	General Purpose LLM (Original)	1.8B, 7B, 14B, 72B	Dense	Tongyi Qianwen*

c) Deepseek

Open-Weight Models Released by DeepSeek

Model Name	Parameters	Release Date	Key Features
DeepSeek-V2	236B (MoE)	Early 2024	Mixture of Experts architecture, 128K context
DeepSeek-V2-Lite	16B	Early 2024	Lightweight version of V2
DeepSeek-Coder-V2	236B (MoE)	Mid 2024	Specialized for coding tasks
DeepSeek-V3	671B (MoE)	Late 2024	Latest flagship model, 128K context
DeepSeek-R1	671B	Jan 2025	Reasoning-focused model with chain-of-thought
DeepSeek-R1-Distill-Qwen-1.5B	1.5B	Jan 2025	Distilled version

Model Name	Parameters	Release Date	Key Features
DeepSeek-R1-Distill-Qwen-7B	7B	Jan 2025	Distilled version
DeepSeek-R1-Distill-Qwen-14B	14B	Jan 2025	Distilled version
DeepSeek-R1-Distill-Qwen-32B	32B	Jan 2025	Distilled version
DeepSeek-R1-Distill-Llama-8B	8B	Jan 2025	Distilled version
DeepSeek-R1-Distill-Llama-70B	70B	Jan 2025	Distilled version

2)Hallucination check for the below prompt

Write a 4-line poem where every line ends with a word that strictly rhymes with the word ‘Silver’.

a)qwen ai

The winter moon, a pale, cold **sliver**,
Beside the grazing little **chilver**,
Reflected back that
shining **sliver**,
To charm the sleepy, woolly **chilver**.

*Note on the rhyme constraint: "Silver" is famously one of the most difficult words to rhyme in the English language. Words like "shiver," "quiver," or "river" do **not** strictly rhyme with it, as they lack the /l/ consonant sound immediately following the stressed vowel. Therefore, **sliver** and **chilver** (an English word for a ewe lamb) are the only two valid strict rhymes available to fulfill your exact constraint!*

b) Chatgpt

There aren't enough standard English words that **strictly rhyme** with "silver" to write four natural lines. In fact, "**silver**" is famously considered to have no common perfect rhyme in English.

I can instead write a 4-line poem using **near-rhymes** such as *shiver*, *river*, *giver*, and *quiver* if you'd like.

c)Deepseek

Finding a perfect rhyme for "silver" is famously difficult, as there are no common perfect rhymes. Using ****near rhymes (assonance)****, here is a 4-line poem:

****The Silver Wish****

I held a gleaming coin of ****silver****,
Its surface like a moonlit ****river****,
I wished upon its polished ****shiver****,
And felt my heart begin to ****quiver****.

Note: "River," "shiver," and "quiver" are near-rhymes (assonant rhymes) with "silver," as no perfect rhyme exists in the English language.

d)Gemini

In standard English, silver is a famously near-orphan rhyme—words commonly cited like *sliver*, *shiver*, or *river* fail as strict rhymes because they lack the medial /l/ consonant.

To achieve a strict phonological rhyme (/ˈsɪlvər/), the verses below draw on genuine English dialect words, compound forms, and proper nouns:

Across the downs he drove the bleating chilver,
Through heavy mists that make the traveler dilver,
Past running streams of flashing, cold quicksilver,
To find the mountain home of shepherd Wilver.

- chilver (/ˈtʃɪlvər/) — An archaic/dialect English noun for a ewe lamb.
- dilver (/ˈdɪlvər/) — An East Anglian dialect verb meaning to exhaust, daze, or weary.
- quicksilver (/ˈkwɪk, sɪlvər/) — The native English term for liquid mercury.
- Wilver (/ˈwɪlvər/) — A recognized English proper name (notably the given name of baseball Hall-of-Famer Wilver "Willie" Stargell).

3) From open router take the model details with Context window for any 10 models from 3 service providers

<https://openrouter.ai/models>

#	Provider	Model	Context Window
1	OpenAI	GPT-6 Astra	1.1M tokens
2	OpenAI	GPT-6 Astra Pro	1.1M tokens
3	OpenAI	GPT Image 2.5 Sunburst	400K context
4	Anthropic	Claude Opus 5	1M tokens
5	Anthropic	Claude Sonnet 5	1M tokens
6	Anthropic	Claude Fable 5.1	1M tokens
7	Anthropic	Claude Opus 4.8	1M tokens
8	Anthropic	Claude Haiku 4.5	200K tokens
9	Google	Gemini 3.8 Flash	1.05M tokens
10	Google	Gemini 3.8 Flash (batch)	1.05M tokens

Reference Links

<https://ollama.com/library/gpt-oss:120b>

<https://developers.openai.com/api/docs/models/gpt-oss-120b>

<https://openrouter.ai/models>

<https://platform.openai.com/tokenizer>